

COMPUTATIONS AND ML FOR SURJECTIVE RATIONAL MAPS

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ABSTRACT. The present note studies *surjective rational endomorphisms* $f : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ with *cubic* terms and the indeterminacy locus $I_f \neq \emptyset$. We develop an experimental approach, based on some Python programming and Machine Learning, towards the classification of such maps; a couple of new explicit f is constructed in this way. We also prove (via pure projective geometry) that a general non-regular cubic endomorphism f of $\mathbb{P}^2$ is surjective if and only if the set I_f has cardinality at least 3.

1. INTRODUCTION

1.1. The set-up. Let X be a complex projective variety and $f : X \dashrightarrow X$ its rational endomorphism. Denote by $I_f \subset X$ the (closed) indeterminacy locus of f . Then f is called *surjective* if the induced morphism $X \setminus I_f \rightarrow X$ is onto. Such maps were introduced and studied — from the algebro-geometric point of view — in the paper [13] (with X being a projective space $\mathbb{P}^n$). The initial motivation for this study came from the optimal control theory and mathematical genetics. Later a relation of surjective rational maps with algebraic statistics and quantization has also been realized (see [9] and [11]). Another classical subject here is holomorphic dynamics: the given f is very much akin to a *polynomial-like map* (see [3, Ch. 2]); hence one could try to define dynamical degrees of f (with respect to some Kähler form on X) and (re)prove standard relations between them and the entropy (cf. [13, Remark 1.2]).

In the present note, we concentrate on the behavior of surjective maps in families, aiming to show that for certain rational maps the surjectivity is a *generic property*. Namely, let us take $X = \mathbb{P}^2$, so that in projective coordinates any of its rational endomorphisms looks like this:

$$f = [f_0 : f_1 : f_2] : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2,$$

where f_i are mutually coprime homogeneous polynomials of some degree d . These polynomials span a plane Π in the linear system $|\mathcal{O}_{\mathbb{P}^2}(d)|$ of all degree d curves on $\mathbb{P}^2$.¹⁾ Note that $I_f \subset \mathbb{P}^2$ is given by equations $f_0 = f_1 = f_2 = 0$. Then the morphism $f : \mathbb{P}^2 \setminus I_f \rightarrow \mathbb{P}^2$ can be defined in the following coordinate-free way (*Kodaira's construction*): for any point $\mathbf{x} \in \mathbb{P}^2 \setminus I_f$, consider the pencil $\ell \subset \Pi$ of all linear combinations of f_i , which vanish at $\mathbf{x}$; now identify $\mathbb{P}^2$ with the dual plane Π^* of lines and put

$$f : \mathbf{x} \mapsto \ell \in \mathbb{P}^2 = \Pi^*.$$

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¹⁾All notions and facts from algebraic geometry used in this paper can be found in e.g. [4].

So the term *generic*, when applied to the map f , simply means that the corresponding plane $\Pi \subset |\mathcal{O}_{\mathbb{P}^2}(d)|$ is such (or, if being more formal, one may state genericity in terms of points on the relevant Grassmannian).

Further, the 0-scheme I_f varies together with f , of course, and one can always achieve $I_f = \emptyset$ for appropriate f_i . Hence the map f becomes *regular* and its surjectivity is straightforward. Thus, in order to have a fruitful problem here, let us fix a (generic) set I_f , consider the maximal linear system $\Lambda \subset |\mathcal{O}_{\mathbb{P}^2}(d)|$ with the base locus $\text{Bs}(\Lambda) = I_f$, and treat only those planes Π , which belong to Λ . In this setting, the generality problem has been solved completely in [15] for the *quadratic* case, i.e. when $d = 2$. More precisely, [15, Theorem 2] asserts that for every given cardinality $\#I_f$ the set of all $\Pi \subset \Lambda$, defining surjective maps, is *Zariski open* (including the set $\emptyset$).

Presently, we consider the next, *cubic*, case of $d = 3$. Our aim is two-fold. Firstly, we extend the main result of [15] by showing that all such surjective f , with fixed generic I_f , form an open set (see Section 2 below). Secondly, we develop an experimental approach towards a complete description of (cubic) surjective rational maps, which also yields new *explicit* examples of them (see Section 3).

Remark 1.2. Surjectivity property seems like another *robust* substitute for the term “generic” among many others that one comes across in classical algebraic geometry. For instance, given a smooth conic $(q = 0) \subset \mathbb{P}^2$ the set of triples of linear forms l_i , satisfying $q = \sum_i l_i^2$, can be compactified into the *variety of self-polar triangles* — this is a codimension 3 linear section of the Grassmannian $G(3, 5)$, — and the self-polarity condition can be effectively verified (see [4, 2.1.3]). Another example is the non-gradient property for a given quadratic endomorphism of $\mathbb{P}^3$ in terms of smoothness of the Jacobi surface (see [8]). Coming back to our setting, a rational map $f : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ is surjective iff there is no pencil $\ell \in \Pi^*$ as above, with the set-theoretic equality $\text{Bs}(\ell) = \text{Bs}(\Pi)$ (we will call such ℓ an *unruly pencil*). The latter property can be also checked explicitly via the direct calculation (cf. the code in 3.1 below).

1.3. del Pezzo surfaces. For what follows, let us fix I_f , $\Lambda \subset |\mathcal{O}_{\mathbb{P}^2}(3)|$ and $\Pi \subset \Lambda$ as in 1.1. We also assume I_f to be *generic*. In particular, we have $I_f = \{P_1, \dots, P_{9-\delta}\}$ for some $\delta \in \mathbb{Z}$, where the points P_i are in general position (no three on a line and no six on a conic); it is easy to see via the parameter count that $2 \leq \delta \leq 8$ (see 2.1 below).

Now consider the blow-up $\varphi : X \rightarrow \mathbb{P}^2$ of all the P_i . Then X is the celebrated del Pezzo surface of degree δ (see [4, Ch. 8]). It is well-known that $\delta \geq 3$ iff the linear system $|-K_X|$ defines the *anticanonical embedding* $X = X_\delta \subset \mathbb{P}^\delta = \Lambda^*$ of X as a degree δ surface. This allows one to treat the map f , for $\delta \geq 3$, as a linear projection of X onto the plane $\mathbb{P}^2 = \Pi^*$ (cf. Figure 2 below). Then the surjectivity of f is established by a simple geometric argument (see Theorem 2.6). In the remaining $\delta = 2$ case, there is an explicitly constructed unruly pencil ℓ , so that f is not surjective (see Corollary 2.3).

Remark 1.4. One interesting fact about smooth del Pezzo surface X with $\delta \geq 3$ is that its *universal torsor* admits an embedding into a homogeneous space G/P . Here G a semisimple algebraic group with parabolic subgroup P corresponding to the root system of rank $9 - \delta$ in the Picard lattice $\text{Pic}(X)$ (see [18]). It would be interesting to contemplate this phenomenon in terms of the above surjectivity result for f .

Surjective rational maps and del Pezzo surfaces have already interacted with each other (see [12]). Let us make one step further and indicate a relation between (surjective) cubic maps f and *rational elliptic surfaces*. First of all,

recall that to an arbitrary surjective endomorphism $f : \mathbb{P}^n \dashrightarrow \mathbb{P}^n, n \geq 1$, one associates a vector bundle $\mathcal{E} \simeq \Omega_{\mathbb{P}^n}^1$ together with an inclusion $f^*(\mathcal{E}) \subseteq \mathcal{E} \otimes \mathcal{O}_{\mathbb{P}^n}(-d+1)$, where f^* is a properly defined pull-back (see [9]). Thus one may regard every such f as a certain connection ∇ on the (“Schwarzenberger-like”) bundle $\mathcal{E}$ (cf. [4, 2.2.2]). Secondly, in our case of $n = 2, d = 3$ projectivization $\mathbb{P}(\mathcal{E})$ is identified (via f) with a family of *elliptic pencils* ℓ , which yields a *flat family* of rational elliptic surfaces $\mathfrak{X} \rightarrow \mathbb{P}^2$. This $\mathfrak{X}/\mathbb{P}^2$ resembles one of the *Painlevé families* (see e.g. [17], [7]), where now compatibility with ∇ plays the role of the isomonodromy condition.

In view of the present discussion, it is tempting to look for (hidden) symmetries associated with f , which one may construct in terms of sublattices in $\text{Pic}(X)$ for the fibers $X \subset \mathfrak{X}$ (cf. Remark 1.4). Such instances as modular transformations, or q-mutations as their quantum analog (see [19]), plus various types of dualities and correspondences (like the S-duality in the del Pezzo case — see e.g. [14]), are awaiting for their way towards surjective rational maps.²⁾ Finally, we can not help mentioning the *spectral data* approach in this context (cf. [6]), which also has some interactions with surjectivity and quantization (see [11]).

1.5. Experiments. The use of various neural networks is so ubiquitous in today’s scientific research that it is difficult to bypass this subject. Presently, we have developed an experimental approach, whose ultimate goal is an effective description/classification of all surjective rational endomorphisms of $\mathbb{P}^2$ (not necessarily having $d = 3$). It goes as follows. First one inspects all dominant maps $\mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ over a small finite field $\mathbb{F}_p$ and generates a table (aka data set) by the rule (see 3.1 below): $f \mapsto \mathbf{1}$ (surjective) or $\mathbf{0}$ (not surjective). After this one learns the table via NN and generates a function $f \mapsto \Psi(f)$ (a “measure of surjectivity”), now defined for *all* f over $\mathbb{C}$, aiming to predict whether a given map is surjective or not (see 3.4).³⁾ Note that this line of thought is not new and resembles the one of *Natural Intelligence* as demonstrated in the wonderful book [1]. There, a lot of numerical experiments over small $\mathbb{F}_p$ ’s, concerning finite dynamics, statistics of discrete functions and finite projective structures, were carried out (by hand!), after what fruitful conjectures got formulated and even proved in some cases.

Further, as is typical to the experiments, their output is often confusing and should be taken with some proviso. Our small project is not an exception. The major issue is that the function Ψ does not take *integer* values and hence it is not clear, given the value $\Psi(f)$, how “far” the map f is from being surjective (this puzzle creates, obviously, a groundwork for future research). Furthermore, the behavior of Ψ shows that the openness property of surjective cubic maps, discussed earlier in 1.3, *does not* hold over $\mathbb{F}_2$. Still, the outcome of our calculations was not empty, and we managed to construct a couple of non-trivial surjective planar maps with $d = 3$; here we could not completely avoid the usage of NI (see Propositions 3.2 and 3.6, as well as the whole Section 3 for further discussion and details).

In conclusion, let us refer to the survey [5] for an excellent overview of how the Machine Learning applies to various questions of pure science, including extremal Calabi–Yau metrics, knots and string theory. We hope that surjective rational maps will make their path to these advanced subjects in future.

²⁾Another example of how the geometry of del Pezzo surfaces can be applied to (wild) automorphisms of threefolds is contained in [10].

³⁾One may regard this reasoning as a kind of “numerical Hasse principle”. It is worth to compare this principle with various (*topological*) *persistence* phenomena that one encounters in applied sciences (see e.g. the survey [2]).

2. SURJECTIVITY RESULTS

2.1. The case $\delta \leq 2$. Note that passing through the (general) points $P_1, \dots, P_{9-\delta}$ as above imposes $9 - \delta$ independent linear conditions on the linear system $|\mathcal{O}_{\mathbb{P}^2}(3)| \simeq \mathbb{P}^9$. Hence we get $0 \leq \delta \leq 9$. Furthermore, if $\delta \leq 1$, then $\dim \Lambda < 2$ and there is no plane $\Pi \subset \Lambda$.

Now let us assume that $\delta = 2$. Then we obtain $\Pi = \Lambda$ and the following holds:

Proposition 2.2. *There exists an unruly pencil $\ell \subset \Pi$.*

Proof. Note that by the parameter count as above there is a cubic $C \in \Pi$, which has P_1 as its inflection point. In turn, this C is contained in a pencil ℓ of cubics from Π , sharing a common tangent with C at P_1 . Take some $Z \in \ell \setminus \{C\}$. Then we have

$$C \cap Z = \{P, P_1, P_2, \dots, P_7\}$$

for some point P . Now, if $P \neq P_1$, then again the parameter count gives $\dim \ell = 9 - 6 - 2 - 1 = 0$ (“1” is for passing through extra P), a contradiction. Hence $P = P_1$ (see Figure 1) and ℓ is unruly.

FIGURE 1. Two curves C and Z

□

From Proposition 2.2 and Remark 1.2 we deduce the following:

Corollary 2.3. *Generic rational endomorphism f of the plane, having cubical components and $\#I_f = 9 - \delta \geq 7$, is not surjective.*

2.4. The case $\delta \geq 3$. Let us resolve indeterminacies of the map $f : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ via the blow-up of I_f ; this is possible due to the generality assumption (cf. **1.3**):

(2.5)

$$\begin{array}{ccc} & X & \\ \varphi \swarrow & & \searrow \psi \\ \mathbb{P}^2 & \xrightarrow{\quad f \quad} & \mathbb{P}^2 \end{array}$$

Here φ is the blow-up of $P_1, \dots, P_{9-\delta}$, so that the birational transform $\varphi_*^{-1}(\Lambda)$ coincides with the linear system $|-K_X|$ for the canonical class $K_X = \varphi^*(K_{\mathbb{P}^2}) + E_1 + \dots + E_{9-\delta}$, where $E_i := \varphi^{-1}(P_i)$ are the exceptional divisors. It follows that ψ coincides with a *regular* projection of the degree δ smooth surface $X = X_\delta \subset \mathbb{P}^\delta = \Lambda^*$ onto the plane $\mathbb{P}^2 = \Pi^*$.

Let us fix a point $o \in \mathbb{P}^2$. In order to show that $f^{-1}(o) \neq \emptyset$ it suffices to check the condition $\psi^{-1}(o) \not\subset E_i$ for all $1 \leq i \leq 9 - \delta$ (cf. [15, Proposition 3]). Here is the main result of this section:

Theorem 2.6. *In the previous setting, we have $\psi^{-1}(o) \not\subset E_i$ for all i and o , so that the map f is surjective.*

Proof. Let $\Sigma \subset \mathbb{P}^\delta$ be the center of the projection ψ — this is a subspace of codimension 3 and ψ is given by the linear system of all hyperplanes passing through Σ . In particular, since $E_i \subset X$ is a line, its image $\psi(E_i) \subset \mathbb{P}^2$ is also a line. Then we have $\psi^{-1}\psi(E_i) = X \cap H = E_i + R$ for some hyperplane H and a curve $R \sim -K_X - E_i$. It follows from the generality assumption on f that H is a generic one. Then we claim the following:

Lemma 2.7. *R is smooth.*

Proof. Consider the projection $\mathbb{P}^\delta \dashrightarrow \mathbb{P}^{\delta-2}$ from the line E_i . This rational map can be regularized via the single blow-up $\nu : W \rightarrow \mathbb{P}^\delta$ of E_i . We have $\nu_*^{-1}(X) \simeq X$ and the restriction of the obtained morphism $W \rightarrow \mathbb{P}^{\delta-2}$ to $X \subset W$ coincides, by construction, with the one given by the linear system $|R|$. Hence we get $\text{Bs}(|R|) = \emptyset$. Then R is smooth by Bertini's theorem and generality of H . $\square$

Further, note that $R \cdot E_i = 2$, which means $R \cap E_i = \{o_1, o_2\}$ for some (distinct) points o_j . Now that ψ maps the union $R \cup E_i$ onto $\psi(E_i)$, in order to prove our Theorem it suffices to show the morphism $\psi|_R$ is *not ramified* at o_1, o_2 (cf. Lemma 2.7 and Figure 2). But this is immediate — again by generality of f (and Σ).

Thus we obtain $\psi^{-1}(o) \not\subset E_i$ for all i and o . Theorem 2.6 is proved.

FIGURE 2. Projection onto $\psi(E_i)$

$\square$

3. EXPERIMENTS AND EXAMPLES

3.1. Quartic case. Here we consider a particular instance of (general cubic) surjective maps f with the locus I_f consisting of five points in general position. According to the reasoning outlined in Section 2 such maps correspond to certain projections $X \rightarrow \mathbb{P}^2$ of a del Pezzo surface $X \subset \mathbb{P}^4$ of degree $\delta = 4$.

Specifically, let us form a linear system $\Lambda \subset |-K_{\mathbb{P}^2}|$ of all cubics on $\mathbb{P}^2$, which pass through the points $[1 : 0 : 0]$, $[0 : 1 : 0]$, $[0 : 0 : 1]$, $[1 : 1 : 1]$, $[2 : 3 : 1]$. Ideally, given a plane $\Pi \subset \Lambda \simeq \mathbb{P}^4$, one wants an *explicit* way of checking whether the corresponding map $f : \mathbb{P}^2 \dashrightarrow \Pi^*$ is surjective. In other words, one wants to exclude

(explicitly) all unruly pencils in Π , as we have discussed in Remark 1.2. The most straightforward way is to restrict everything to a finite field $\mathbb{F}_p$ and list all the possibilities. Here is a Python-Sage code doing this:⁴⁾

```

from sage.all import *

import numpy as np

# specify a finite field and algebraic data over it

k = GF(Integer(2)); V = VectorSpace(k,5); VV = VectorSpace(k,3)

R = k['x', y, z]; (x, y, z) = R._first_ngens(3)

PP = ProjectiveSpace(Integer(2), k, names=('x', 'y', 'z'));
(x,y,z) = PP._first_ngens(3)

# ideal defining the set of points [1:0:0], [0:1:0], [0:0:1],
# [1:1:1], [2:3:1]

I = ideal(x^2*y + 3*y^2*z - x*z^2 - 3*y*z^2, x*y^2 + 3*y^2*z -
2*x*z^2 - 2*y*z^2, x^2*z - y^2*z + 2*x*z^2 - 2*y*z^2, x*y*z +
3*x*z^2 + 3*y*z^2,
          z*(x - y)*(y - x - z))

B = I.groebner_basis(deg_bound=Integer(3))

w = vector(B)

# compute number of irreducible component in a scheme s

def irred(s):
    ir_comp = s.irreducible_components()
    return len(ir_comp)

# create P2 in the given subspace of cubics

```

⁴⁾To lighten the calculations we consider Λ over $\mathbb{F}_7$ (cf. generators of the ideal I below).

```

def make_plane(v, u, t):
    p = ideal(v.dot_product(w), u.dot_product(w), t.dot_product(w))
    return p

# number of base points in a given P2

def base_locus_length(v, u, t):
    some_plane = make_plane(v, u, t)
    base_loc = PP.subscheme(some_plane)
    return irred(base_loc)

# create a pencil in the given P2

def make_pencil(v, u, t, a, b):
    s_plane = make_plane(v, u, t)
    c_base = s_plane.groebner_basis(deg_bound=Integer(3))
    v_tran = vector(c_base)
    penc = ideal(a.dot_product(v_tran), b.dot_product(v_tran))
    return(PP.subscheme(penc))

# for given P2 inspect over all its pencils and return 0 if base

# loci of P2 and a pencil in it have the same length

def num_surjective(v, u, t):
    x = 0
    for a in VV:
        for b in VV:
            vari = make_pencil(v, u, t, a, b)
            if vari.dimension() > 0:
                continue
            else:
                if irred(vari) == base_locus_length(v, u, t):
                    return 0
                return
            else:
                x = 1
                continue
    return x

```

```

# auxiliary function: sum of squares of coordinates

def sq_coord(vect):
    s = sum(x**2 for x in vect)
    return s

# auxiliary function: check whether a collection of three vectors

# is orthonormal => restrict to such v, u, t below

def is_orth(vectors):
    for i in range(3):
        if sq_coord(vectors[i]) != 1:
            return False
    for j in range(i + 1, 3):
        if np.dot(vectors[i], vectors[j]) != 0:
            return False
    return True

# list all dominant maps and print them into file

dic_val = {}

for v in V:
    for u in V:
        for t in V:
            var = PP.subscheme(make_plane(v, u, t))
            if matrix([v, u, t]).rank() < 3 or var.dimension() > 0:
                continue
            else:
                if __name__ == "__main__":
                    vectors = np.array([tuple(v), tuple(u), tuple(t)])
                    if is_orth(vectors) == False:
                        continue
                    else:
                        dic_val[tuple(v), tuple(u), tuple(t)] =
                            num_surjective(v, u, t)

```



```

filename = 'output.txt'

with open(filename, 'w') as file:
    for key, value in dic_val.items():
        file.write(f"{key}: {value}\n")

print(f"Values have been written to {filename}")

```

The program lists all vectors $\mathbf{v}$, $\mathbf{u}$, $\mathbf{t}$ from Λ , spanning various planes Π as above, and writes the result to the file `output.txt` in the form $(\mathbf{v}, \mathbf{u}, \mathbf{t}): \text{value}$. Here `value` equals either `1` or `0`, depending on whether the corresponding map f is surjective or not, respectively.

To reduce the amount of calculations we inspect only those vectors $\mathbf{v}$, $\mathbf{u}$, $\mathbf{t}$ that form an orthonormal basis in Π (cf. the function `is_orth`). However, even under this restriction our program was able to generate `output.txt` only for $p = 2$, as for larger p the computation becomes unreasonably long.⁵⁾ But still, these experimental results over $\mathbb{F}_2$ allow one to obtain non-trivial examples of surjective f , now defined over $\mathbb{C}$.

Namely, let us turn to the case no. **46** in `output.txt`, with $(\mathbf{v}, \mathbf{u}, \mathbf{t})$ being $((1, 0, 0, 0, 0), (0, 0, 0, 1, 0), (1, 1, 0, 0, 1))$ and `value` equal `1`. The linear system Λ is spanned by the cubics

$$x^2y + y^2z, xy^2 + y^2z, x^2z + y^2z, xyz, xz^2 + yz^2$$

(cf. vector $\mathbf{w}$ in the above code) — please mind that all this is over $\mathbb{F}_2$.⁶⁾ Now let us (formally) extend Λ with the present basis to be defined over $\mathbb{C}$.

Thus we are considering $f: \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ given by

$$[x : y : z] \mapsto [x^2y + y^2z : xyz : x^2y + xy^2 + 2y^2z + z^2(x + y)].$$

Note that $I_f = \text{Bs}(\Lambda) = \{[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]\}$, i.e. the points $[1 : 1 : 1]$ and $[2 : 3 : 1]$ got lost in the way, as results from the mod 2 reduction and extension to $\mathbb{C}$ procedure. We claim that

Proposition 3.2. *The given map f is surjective.*

Proof. Put $P := [1 : 0 : 0]$. Then it is immediate that $f^{-1}(P) \ni [0 : 1 : -2]$. Next we take $Q := [0 : 0 : 1]$ and observe that $f^{-1}(Q) \ni [1 : 0 : 1]$.

Further, consider the line $L = (y = 0) \ni P, Q$ and pick an arbitrary point $[1 : 0 : a] \in L \setminus \{P, Q\}$. Then it is immediate that $f([0 : 1 : a - 2]) = [1 : 0 : a]$. Thus the map f is *surjective over L* .

Let us now verify surjectivity over the complement $\mathbb{P}^2 \setminus L$. Any point in the latter is of the form $[a : 1 : b]$ for some a and $b \in \mathbb{C}$. Hence we must have $xyz \neq 0$. Then, letting $z = 1$, it suffices to show that the point $(a, b) \in \mathbb{C}^2$ has a preimage under the map

$$(x, y) \mapsto \left(x + \frac{y}{x}, x + y + \frac{2y}{x} + \frac{1}{x} + \frac{1}{y}\right)$$

⁵⁾In this respect, it is worth developing an algorithm, which will not distinguish between *projectively equivalent* triples $(\mathbf{v}, \mathbf{u}, \mathbf{t})$.

⁶⁾The shape of the components of $\mathbf{w}$ is more complicated when $p \geq 5$.

(we have divided the components of f by xy). The condition $x + \frac{y}{x} = a$ gives $y = ax - x^2$ (with non-zero x) and for the second component we get

$$x^2y + xy^2 + 2y^2 + x + y = bxy.$$

After substituting the expression for y and dividing by x we obtain the equation

$$(3.3) \quad x^2(a - x) + x^2(a - x)^2 + 2x(a - x)^2 + 1 + a - x - bx(a - x) = 0.$$

Note that $x = 0$ forces $a = -1$. Next we bring (3.3) to the form

$$x^4 + (1 - 2a)x^3 + (a^2 - 3a + b)x^2 + (2a^2 - ba - 1)x + a + 1 = 0.$$

For $a \neq -1$, the latter equation has a non-zero solution, and so (a, b) has a preimage by our construction. Assume that $a = -1$. Then the equation becomes (after dividing by x)

$$x^3 + 2x^2 + (b + 4)x + b + 1 = 0.$$

Again, if $b \neq -1$, then we are done. Finally, in the case of $b = -1$ we get quadratic equation $x^2 + 2x + 3 = 0$, which has a non-zero solution. This completes the proof that f is surjective. $\square$

3.4. Entering ML. The above reasoning aims to illustrate a general strategy for finding non-trivial surjective rational endomorphisms of $\mathbb{P}^2$: make predictions over (small) finite field (cf. file `output.txt`) and then inspect those `v`, `u`, `t`, which “likely give” surjective f (cf. `value = 1` condition). The latter requires, as Proposition 3.2 reveals, quite tedious (although direct) geometric manipulations. For the former, a seemingly important question is:

how to make “predictions of surjectivity” for *arbitrary* vectors `v`, `u`, `t`, not necessarily over $\mathbb{F}_p$?

One way to answer this, as became common nowadays, is to generate a “prediction function” via neural network by inspecting the table from `output.txt` (cf. the discussion in 1.5). Here is the corresponding code:

```
import ast
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from keras.models import Sequential
from keras.layers import Dense, Conv2D, Flatten

# create a dictionary from the given file

def read_dict_file(filename: str) -> dict:
    result = {}
    with open(filename, 'r') as file:
        for line in file:
            key, value = line.strip().split(':')
```

```

        result[ast.literal_eval(key)] = int(value)
    return result

if __name__ == "__main__":
    filename = 'output.txt'
    my_dict = read_dict_file(filename)

# function we are going to learn

def func_from_dict(u, v, w):
    return my_dict[u, v, w]

# prepare data (not scaled)

X_first = np.array(list(my_dict.keys()))

Y = np.array(list(my_dict.values())).reshape(-1, 1)

# scale data

scaler_Y = StandardScaler() Y_train =
scaler_Y.fit_transform(Y_train) Y_test =
scaler_Y.transform(Y_test)

X_list = []

for matrix in X_first:

    matrix_reshaped = matrix.reshape(-1, matrix.shape[-1])
    scaler_X = StandardScaler()
    scaled_matrix = scaler_X.fit_transform(matrix_reshaped)
    scaled_matrix = scaled_matrix.reshape(matrix.shape)
    X_list.append(scaled_matrix)

X = np.array(X_list)

# split into train and test sets

X_train, X_test, Y_train,

```

```

Y_test = train_test_split(X,Y,test_size=0.2, random_state=42)

# build neural network

model = Sequential() model.add(Conv2D(256, kernel_size=(2,2),
activation='relu ', input_shape=(3,5,1))) model.add(Flatten())
model.add(Dense(256, activation='relu ')) model.add(Dense(1))

# compile model

model.compile(optimizer='adam', loss='mean_squared_error')

# model summary

model.summary()

# train model

model.fit(X_train, Y_train, epochs=150, batch_size=32)

# evaluate model loss = model.evaluate(X_test, Y_test)
print(f'test loss: {loss}')
```

```

# make predictions

Y_learn = model.predict(X_test)

Y_learn = scaler_Y.inverse_transform(Y_learn)

print(f'predictions: {Y_learn}')
```

Our model is based on the Keras API. The NN's configuration (one of many possible) includes the input layer with 256 neurons and one hidden layer with 256 neurons. After training the model the pertinent prediction function Y_{learn} is obtained.⁷⁾ Its average against the test data can be found:

```
s = Y_learn.sum()
```

⁷⁾We did not intend to be in any way optimal here and the test loss turns out to be 0.078. We also do not care for any sort of interpretability (but of course, applying KAN neural network, say, in our setting seems very intriguing — see [16]).

```
t = len(X_test)
```

```
print(s/t);
```

and is equal to 0.0737. These calculations reveal a curious fact that over the field $\mathbb{F}_2$ (over which `output.txt` has been generated) the analog of Theorem 2.6 *does not hold*: for average rational map f (aka `(v, u, t)`), its Y_{learn} -value is close to 0.

Thus with *any* f one may associate its *measure of surjectivity* $Y_{\text{learn}}(f)$ and ask to what extent does the latter reflect the corresponding geometric property of the given map. The preceding computer experiments, as well as Proposition 3.2 and its proof (see also 3.5), show that this is a difficult problem in general.

3.5. Cubic case. Let us add one more point $[3 : 2 : 1]$ to the base locus of Λ from 3.1. This corresponds to erasing the term $\mathbf{z}*(\mathbf{x} - \mathbf{y})*(\mathbf{y} - \mathbf{x} - \mathbf{z})$ in the ideal I from the above Python-Sage code.⁸⁾ Now, after running the script (with `V = VectorSpace(k,4)`), the file `output.txt` is generated in a reasonable time again only for $p = 2$. One can see that now `value` equals `0` for all triples `(v, u, t)`. Thus $Y_{\text{learn}}(f) = 0$ for all f , i.e. they are “close to be non-surjective”, according to the heuristic principle in 3.4. Yet, amazingly, one can find surjective maps among these f . Let us proceed as earlier.

Namely, the basis of Λ now consists of cubics

$$x^2y + y^2z + z^2(x + y), \quad y^2(x - z), \quad z(x^2 + y^2), \quad xyz + xz^2 + yz^2,$$

and we consider the triple $((1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 0, 1))$. Then the corresponding map $f : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ is given by

$$[x : y : z] \mapsto [x^2y + y^2z + z^2(x + y) : y^2(x - z) : xyz + z^2(x + y)].^{9)}$$

Again we have $I_f = \text{Bs}(\Lambda)$ (over both $\mathbb{F}_2$ and $\mathbb{F}_7$).

Proposition 3.6. *The given f is surjective.*

Proof. Put $P := [1 : 0 : 0]$. Then it is immediate that $f^{-1}(P) \ni [-2 : 1 : -2]$. Next we take $Q := [0 : 0 : 1]$ and observe that $f^{-1}(Q) \ni [-1 : 1 : -1]$.

Further, consider the line $L = (y = 0) \ni P, Q$ and pick an arbitrary point $[a : 0 : 1] \in L \setminus \{P, Q\}$. Then for $[x : y : z] \in f^{-1}([a : 0 : 1])$ we get the following conditions:

$$x = z = 1, \quad \frac{(y + 1)^2}{2y + 1} = a, \quad y \neq -0.5.$$

This yields quadratic equation $y^2 + 2(1 - a)y + 1 - a = 0$. The latter always has a solution $\zeta \neq -0.5$. In particular, we obtain that $[1 : \zeta : 1]$ is contained in $f^{-1}([a : 0 : 1])$, and hence the map f is *surjective over L* .

⁸⁾Once again everything is over $\mathbb{F}_7$ for simplicity.

⁹⁾Here, as in 3.1, all data is considered over $\mathbb{C}$. Hence we distinguish $y^2(x - z)$ and $y^2(x + z)$ from the initial $\mathbb{F}_2$ -basis of Λ .

Let us now verify surjectivity over the complement $\mathbb{P}^2 \setminus L$. Any point in the latter is of the form $[a : 1 : b]$ for some a and $b \in \mathbb{C}$. Hence we must have $y \neq 0$. Then, letting $y = 1$, it suffices to show that the point $(a, b) \in \mathbb{C}^2$ has a preimage under the map

$$(x, y) \mapsto \left(\frac{x^2 + z + xz^2 + z^2}{x - z}, \frac{xz + xz^2 + z^2}{x - z} \right)$$

or, after transforming $(a, b) \mapsto (a - b, b)$, under the map

$$(x, y) \mapsto \left(\frac{x^2 + z - xz}{x - z}, \frac{xz + z + xz^2 + z^2}{x - z} \right).$$

The condition $\frac{x^2 + z - xz}{x - z} = a$ gives $z = \frac{ax - x^2}{1 + a - x}$ (with $x \neq 1 + a, z$), and for the second component we get

$$xz + z + xz^2 + z^2 = b(x - z).$$

Now, substituting, dividing by x and clearing denominators gives the equation

$$(3.7) \quad x^4 + (-2a + 2)x^3 + (a^2 - 4a)x^2 + (2a^2 - a + b - 1)x + a^2 + a(1 - b) - b = 0.$$

Note that the condition $z = \frac{ax - x^2}{1 + a - x} = x$ forces $x = 0$. Quartic polynomial in (3.7) has only this value as its root iff it coincides with x^4 . This gives two incompatible relations $2a - 2 = 0 = a^2 - 4a$ (for the coefficients at x^3 and x^2). Hence we have verified that $x \neq z$.

It remains to deal with the case when possibly $x = 1 + a$. After substituting in (3.7) one gets $a^2 + 3a + 2 = 0$. On the other hand, from the expression $\frac{x^2 + z - xz}{x - z} = a$, with $x = 1 + a$, we obtain $a = -1$, a contradiction. This completes the proof that f is surjective. $\square$

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